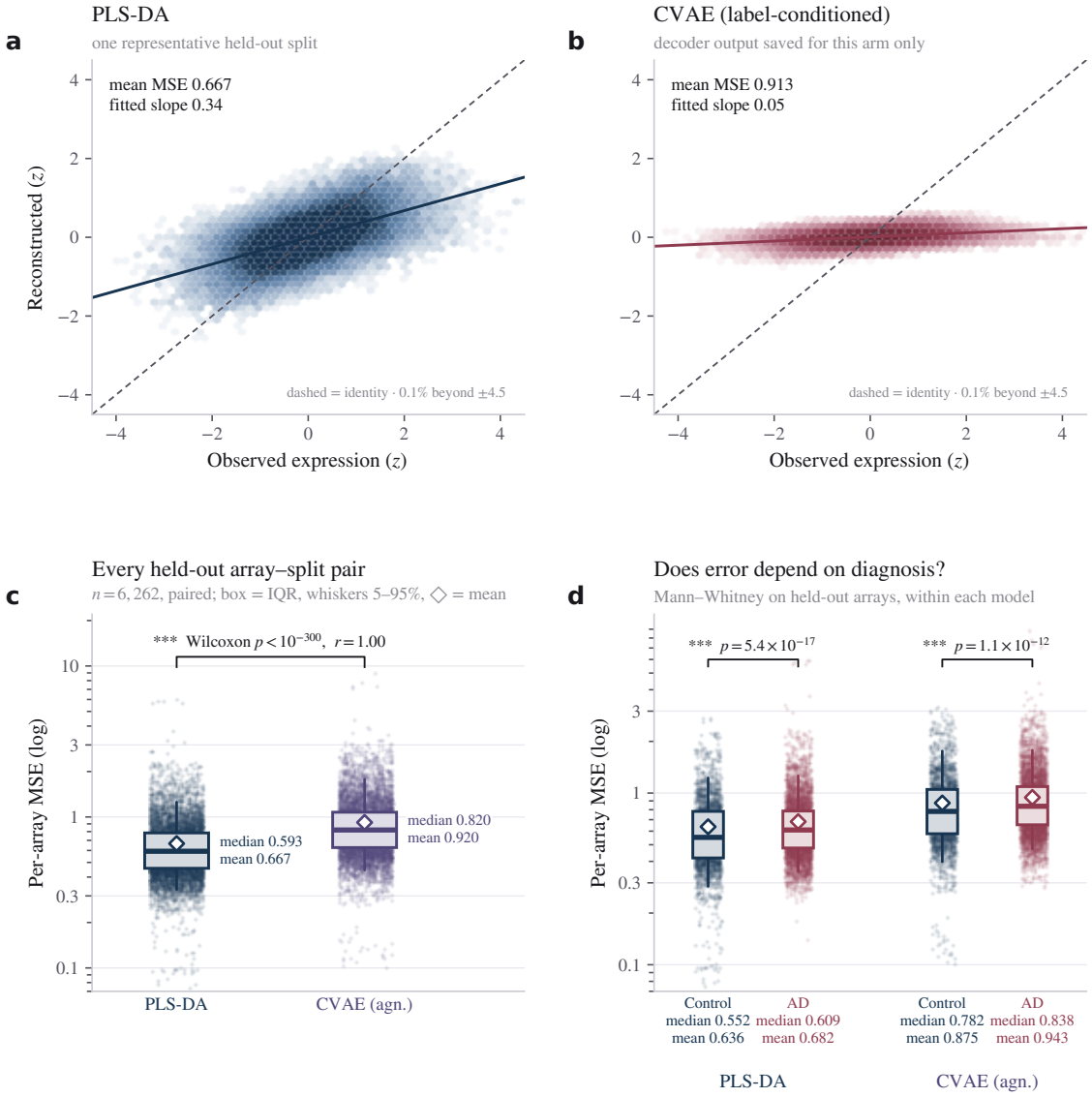

a
PLS-DA
one representative held-out split
mean MSE 0.667
fitted slope 0.34
dashed = identity · 0.1% beyond ±4.5
Reconstructed (z)
Observed expression (z)
b
CVAE (label-conditioned)
decoder output saved for this arm only
mean MSE 0.913
fitted slope 0.05
dashed = identity · 0.1% beyond ±4.5
Observed expression (z)
c
Every held-out array–split pair
n = 6, 262, paired; box = IQR, whiskers 5–95%, ◇ = mean
*** Wilcoxon p < 10−300, r = 1.00
Per-array MSE (log)
median 0.593
mean 0.667
median 0.820
mean 0.920
PLS-DA
CVAE (agn.)
d
Does error depend on diagnosis?
Mann–Whitney on held-out arrays, within each model
*** p = 5.4 × 10−17
*** p = 1.1 × 10−12
Per-array MSE (log)
Control
median 0.552
mean 0.636
AD
median 0.609
mean 0.682
Control
median 0.782
mean 0.875
AD
median 0.838
mean 0.943
PLS-DA
CVAE (agn.)